Individual Differences in Mature Readers in Reading, Spelling, and Grapheme-phoneme Conversion

JOHN R. BEECH
University of Leicester, UK

Individual differences in word recognition, spelling, verbal reasoning, frequency of print exposure, phonological skills, Gestalt processing and semantic processing were examined in 110 university undergraduates. Most reading and phonological tasks involved making sequences of lexical decisions through lists of common stimuli within set time limits. The test measures were percentages correct adjusting for chance. The main findings were as follows: First, there is a substantial "Matthew effect" in that good readers were also good at spelling, good at handling the meaning of words, and appeared to have had the most print exposure. Second, phonological skills appear to be independent of lexical skills. Furthermore, within these phonological skills there appear to be two components, one that is automatic, and one that is more of a conscious operation.

INTRODUCTION

There has been little research into individual differences in processes involved in word recognition and spelling ability in mature readers compared with children. There have been efforts in the past to classify mature readers into specific categories (e.g., Baron & Strawson, 1976). However, there has been little work examining variations in normal readers to see whether differences in modular skills, such as in the hypothesised grapheme-phoneme converter (GPC), could have a functional impact on overall reading and spelling ability.

One question to explore is whether having well-developed decoding skills will be associated with a developed reading vocabulary in later years. It seems that there is a symbiotic relationship between proficiency in GPC and the development of lexical reading skills in the early years of reading. For instance, there is much work establishing that phonological differences in early readers have an impact on subsequent reading development (e.g., Bradley & Bryant, 1978; Wagner & Torgesen, 1987). It has been argued that reading improves phonological skills and these in turn assist the development of sight vocabulary. For example, Morais et al. (1979) found that Portuguese non-readers who learned to read in adulthood had phonological awareness, which was not present in non-readers of the same social background who continued to be illiterate. It might follow that mature readers who have good decoding skills (within the GPC) should also be good

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at lexical processing in the visual mode. Baron and Strawson (1976) selected what they called "Chinese" and "Phonecian" readers, based on performance in GPC versus lexical performance. The Phonecians were chosen as being good on a test deciding if non-words sounded like real words (e.g., *caik* vs. *burb*), but weak on a lexical test. By contrast, the Chinese readers were weak on the induced GPC task, but good on the lexical task. They found a weak non-significant correlation between measures of these two processes across an undifferentiated pool of subjects. This suggests that those good in lexical processing are not necessarily good in their GPC skills. A related consideration is that grapheme-phoneme correspondence skills might divide into two different kinds of skill. On the one hand there is the conscious GPC as measured in the Baron and Strawson induced GPC task. Nonwords have to be read and a phonological code consciously generated (*/kAk/* for *cake* – notation of Beech, 1989). This code is then used to access the auditory lexicon. On the other, there could be a more automatic GPC outside conscious control that operates as checking function. This could be the kind of process that functions in tasks such as those involving comparing regularly and irregularly spelled words. Where there are differences, these would be induced by the rapid automatic processing of grapheme-phoneme codes. These two aspects of GPC functioning may or may not be related to one another.

The measures under consideration included word recognition, spelling, verbal reasoning, print exposure, phonological skills, Gestalt processing and semantic processing. Word recognition was assessed by presenting words within a lexical decision task. The words varied in frequency of usage. Those less skilled in reading would be less familiar with the lower frequency words. All the words were irregular in spelling. Thus, in order to be good at this task one must have good orthographic knowledge of a large sample of English words. The author recognition test, on the basis of previous research (Stanovich & West, 1989), should correlate with this measure of word recognition skill. This would confirm that skill in lexical access is produced by such environmental factors as frequency of exposure. Similarly, spelling ability should be correlated with reading ability. Those who are more familiar with orthography, in terms of reading exposure, should be correspondingly better in spelling. By contrast, Olson et al. (1989) have argued from studies of identical and fraternal twins that phonological coding, particularly the deficit found in reading disabled children, has a genetic component. Phonological skills were studied in the present study by means of a variety of measures as elaborated in the method section. The measure of reasoning ability was taken as a means of extracting variance that could be attributable to a measure of verbal intelligence. The last two measures, Gestalt processing and semantic processing, are two other factors that could have an influence on the skill of word recognition.

METHOD

Subject

The participants were 110 undergraduate students studying psychology at the University of Leicester who volunteered as part of their experimental participation requirement. There were 90 females and 20 males taking part. The mean age was 20.2

years (4.4 years SD). The ages ranged from 18 to 47 years and the distribution was 35.5 percent, 23.7 percent, 19.1 percent, 5.5 percent, and 12.6 percent for those of 18, 19, 20, 21, and those over 21 years, respectively.

Design and Procedures

All testing was performed under conditions of complete silence in a group setting with approximately 15 to 20 individuals in each group. The experiment was a repeated measures design. All the tests were contained in a booklet that was worked through under the supervision of the author, who was positioned so as to see all of them simultaneously.

For each test, participants were told not to begin until instructed to do so. They would hear the word "stop" when the time was up. They had to lift their pens off the paper immediately, but could not turn over to the next page until instructed. In the majority of the tests they had to work down lists of items and mark each as "1" or "2," depending on the nature of the task. The following tasks were exceptions to this: (a) In the Spelling Component Test, they had to fill in the missing letters within the blank spaces. (b) In the Semantic Link Test there was one of four choices for each item. (c) In the Baddeley Reasoning Task they had to write "T" or "F" to indicate true or false.

The experimenter explained what was required before each test was tackled. This included saying that the items were numbered and the direction of working through the items that was required. For instance: "There will be two columns of items; work down each column in turn starting from the left. Don't skip any item."

Each task was scored according to the number correct and was corrected for guessing by subtracting the number of incorrect responses. If this resulted in a negative score, this was set to zero. The final score was adjusted for the total number of items in the test so that all measures produced a potential maximum score of 100.

Stimuli

The tests were usually completed within about 1 to 3 minutes, with the longest taking 10 minutes. These tests were as follows:

Tests of Reading Vocabulary, Print Exposure, Spelling and Reasoning:

1. *Silent Adult Reading Test.* (Beech, 1993). This test of reading vocabulary consists of 84 real words and 42 fake words. The task is to work through the list identifying the real words. This produces a sampled measure of overall reading vocabulary. It was scored by deducting real words that they failed to identify from the real words that they correctly identified. (Duration: 4 min; length 126 items.)
2. *Author Recognition Test* (Stanovich and West, 1989). Participants are presented with what they think is a list of authors and they have to mark those with whom they are familiar. However, half of the items are false names

(devised by the author). This was scored by taking the number of correctly identified real authors and deducting the number of real authors that they failed to identify. The result was multiplied by two to produce a percentage correct. The test appears to be a good indicator of long-term exposure to print. Those who have read a lot of books should be more familiar with the names of authors. (Duration: 3 min; length 100 items.)

3. *The Spelling Component Test.* An incomplete word (e.g., "ac__modate") had to be completed (Coren, 1989). (Duration: 10 min.; length 76 items.)
4. *Baddeley Reasoning Test.* (Baddeley, 1968). An example of an item is: "A is not followed by B – BA. True or False?" Participants had to write a "T" or "F" in the blank provided. This test should correlate with the Spelling Component Test (Coren, 1989). The number of items in which participants responded "true" when it should have been "false" was subtracted from the number of correct "true" responses. (Duration: 2 min; length 64 items.)
5. *Semantic Link Test.* (Beech, 1993). This consists of sets of words, one of which (e.g., "adder") has to be connected to one of a set of four (e.g., "snake, thief, adept, and scarp"). The answer in this instance is "snake." This is completed within a fixed period of time. The items were corrected for guessing by taking the number of correct items and deducting the number of incorrect stimuli divided by 3. This task relates to the Baddeley reasoning task and the spelling component test. The items in the test have been reduced from a larger set. (Duration: 6 min; length 38 items.)

Tests Related to the Decoding and Phonological Skills:

6. *Induced GPC.* A list of nonwords is presented in which half can be pronounced as a real word (e.g., *ishoo* has the same pronunciation as *issue*). Participants identify those that can be pronounced as a real word. This task forces grapheme-to-phoneme conversion. It is similar to tasks involving pseudoword naming (e.g., Snowling, 1985) except that the task is silent and is perhaps slightly more naturalistic in this regard. This measure is calculated by deducting the number of nonwords not portraying real words that were erroneously identified as portraying real words (e.g., *smaw*) from the total number of nonwords portraying real words (e.g., *ishoo*) correctly identified. (Duration: 90 sec; length 64 items.)
7. *Homophonic Nonword Reading*: This consisted of two separately completed lists. In the homophonic nonword list the nonwords sounded like real words (e.g., "werd," "offen"). By contrast, the nonhomophonic nonword list had words that did not correspond with real words when pronounced (e.g., "roosy"). In both words the nonwords are close in visual appearance to real words to reduce the use of a potential pattern-matching strategy. The homophony variable was measured by the following formula in which nonhomophony is denoted by *Nh* and homophony by *H*: $(Nh-H)/Nh$. This means that a high positive score indicates that a person finds reading a

nonhomophonous word much easier and thus is susceptible to the homophony effect. Unlike the induced GPC task, phonological coding is not essential as matching can be achieved on a lexical basis, in which case an homophony effect would not occur. However, if there is recoding to a phonological form when doing this task, interference will be experienced and there should be a positive score. Deep dyslexics have a malfunction in phonological reading and are not influenced by homophonic nonwords (Patterson & Marcel, 1977), suggesting that they are reading holistically and are unaware of the sounds that the words make. Also, Beech and Harris (1997) have shown that young deaf readers are uninfluenced on this task compared with hearing controls matched in reading skills. (Duration: 45 sec for each list; length 62 items in each list.)

8. *Regularity Test*. This comprised two lexical decision task lists presented separately, containing words and nonwords. The words were matched in frequency (Kucera & Francis, 1967) across the two lists and were arranged so that the most frequent were earlier in the lists. Letter string lengths were also matched. One list had regularly spelled words and the other irregularly spelled words. Susceptibility to regularity was measured by $(R - I)/R$, where R is performance in the regular list and I performance in the irregular list. Thus a high score indicates that a person finds reading regular words much easier than irregular words and is susceptible to the regularity effect. Those using a lexical-to-meaning route ought to be unaffected by regularity, whereas those who are affected are using a GPC route. For instance, a phonological dyslexic depended on the appearance of words and could not read nonwords, furthermore she was unaffected by spelling regularity (Temple & Marshall, 1983). Beech and Harris (1997) found that young deaf readers are much less affected by spelling regularity than reading age controls, presumably because of their considerably reduced phonological processing. By contrast, a surface dyslexic studied by Coltheart *et al.* (1983) had greater difficulty with irregular words because she read translating letters to sounds. (Duration: 90 sec. for each of the two lists; length 120 items in each list.)
9. *Rhyme Recognition Test*. In two separately presented lists of words, pairs of words were judged whether or not they rhymed (e.g., "would/stood"). The difference in the two lists lies in the non-rhyming pairs. In the visually similar (VS) lists the non-rhyming pairs have the same orthographic structure (e.g., "hive/give"); but in the visually dissimilar (VD) lists the non-rhyming pairs have a different structure (e.g., "cloak/sulk"). In the case of the rhyming pairs, the pairs did not have visually similar structures (e.g., "balm-farm"). In none of these categories did the word stem match at the same time as the rhyme of the word. Comparison of the phonological codes of both words in either list (VS vs. VD) can be either via the lexical-to-phonology route or by the GPC route. A measure of the relative difference between the two lists ([VS-VD]/VD) indicates interference in the non-rhyming pairs in the visually similar list if the score is low. (Because VS scores

TABLE 1
Intercorrelations among the Principal Variables
(N = 110 in all Cells)

Variable		1	2	3	4	5	6	7	8	9
Age	1									
Spelling	2	.22*								
Baddeley Reasoning	3	.08	.19							
Semantic Link	4	.39**	.22*	.24*						
Silent Adult Reading	5	.30*	.46**	.11	.45**					
Author Test	6	.38**	.43**	.24*	.60**	.54**				
Induced GPC	7	.08	.18	.05	-.01	.17	.03			
Homophony	8	.02	-.11	-.14	-.11	-.13	-.19	.08		
Regularity	9	.11	.11	.14	.01	.14	.04	.13	.26**	
Rhyming	10	.01	.14	.04	.08	.25*	.13	.07	.08	.0
Alternating case	11	-.15	.14	.22*	-.08	.06	.09	.03	.07	.3
Meaning judgements	12	.07	.27**	.32**	.19*	.21*	.35*	.23*	-.02	.1

*$p<0.05$ **$p<0.01$.

are lower in comparison to VD scores). This would show a strong response interference effect between wanting to respond "same" to similar letter strings and being required to respond "different" to different rhymes.

Tests of Gestalt and Semantic Processing:

10. *Mixed Case Test.* This was tested as a lexical decision task. These were the same stimuli as in the irregular list of words in the regularity task, except that the items were in a mixed case (e.g., rHyThM). The case of the letters alternated for each letter with half the words beginning with an upper case letter. These two lists were compared. Those who have a literal representation (or Gestalt) of the orthographic appearance of each word should be the most disrupted (e.g., Pring, 1981). In the analysis the following formula was used: *(Mc-Umc)/Umc* where *Mc* was the mixed case list and *Umc* was the unmixed case. Those susceptible to alternating case would have positive scores. (Duration: 90 sec. for each list; length 120 in each list.)
11. *Semantic Reading Test.* Pairs of words were judged if they were similar in meaning. For instance in the case of "reign/king" and "barren/aristocrat," the first pair are associated, but the second pair are not associated in meaning. Scoring corrected for guessing by means of deducting the number of disassociated pairs erroneously classed as associated from the number of associated pairs that were correctly identified. This task requires being able to link the spelling of homophones with their meaning indicated only by their spelling. A similar type of task was given by Stanovich and West (1989). Their orthographic choice task involved judging pairs (e.g., "currency/ currancy") for correctness. A major difference with this task is that in the semantic reading test the participant has to know that the different meanings of "barren" and "baron."

TABLE 2
Means, Standard Deviations, Minimum and Maximum Scores of the Main Variables

Variable	*Mean*	*SD*	*Minimum*	*Maximum*
Age	20.23	4.4	18	47
Spelling	53.70	13.9	25	84
Baddeley Reasoning	21.53	15.6	0	71.9
Semantic Link	44.18	17.1	0	89.5
Silent Adult Reading	69.94	12.3	26.2	92.8
Author Test	22.16	9.2	4	46
Induced GPC	38.30	16.5	0	71.9
Homophony	.04	.24	-.93	.50
Regularity	-.25	.22	-.80	.21
Rhyming	-.17	.40	-1.3	1.86
Alternating case	-.19	.17	-.68	.37
Meaning judgements	38.15	12.3	10	73.33

RESULTS AND DISCUSSION

Table 1 illustrates the intercorrelations between the main variables in the study and Table 2 shows their means and distributions. Although significant correlations at both the 1 percent and 5 percent levels are indicated, the criterion level of significance should be set to .01, because of the large number of correlations and resultant Type 1 risk in Table 1. Performance on the Silent Adult Reading Test correlates significantly with 6 of these variables and included a correlation of .54 with the Author Recognition Test. This correlation is comparable with a correlation of .60 found by Stanovich and Cunningham (1992) between the Author Recognition Test and the Nelson-Denny Vocabulary subtest, a subtest of the Nelson-Denny Reading test, using an undergraduate sample of 300 students. In this subtest participants read an incomplete sentence and chose the appropriate word or phrase to complete the sentence. Coren (1989) provides norms for 702 Canadian university students on his spelling test with 61.1 percent and 56.7 percent correct for females and males respectively. It can be seen that performance in spelling is slightly worse in the present sample at 53.7 percent.

The interrelationships between the main variables were examined further. As the Baddeley task examined reasoning ability, this was taken as a measure of verbal intelligence. Baddeley (1968) found it to be highly correlated with verbal intelligence. Performance on the Baddeley Reasoning Task was forced into each of the following regressions as the initial variable in order to control for verbal reasoning. Table 3 shows the results of four multiple regressions. In the first regression performance on the Silent Adult Reading Test was the dependent variable and all the variables in Table 1, and gender of the participants, were the independent variables. The final equation produced a multiple R of .60. This indicates associations between reading vocabulary, author recognition, spelling, and the semantic link test suggesting that the major literacy measures are related to frequency of exposure to print. In addition, the minor

TABLE 3
Multiple Regression Analyses

Dependent variable	*Beta*	*R^2 change*	*p*
Silent Adult Reading			
Baddeley Test	-.07	.01	ns
Author Recognition	.283	.28	.000
Spelling	.277	.06	.002
Semantic Link	.217	.03	.025
Rhyming	.161	.03	.04
Spelling			
Baddeley Test	.232	.02	ns
Silent Adult Reading	.321	.19	.000
Author Recognition	.094	.04	.024
Regularity			
Baddeley Test	.113	.02	ns
Alternating case	.274	.09	.002
Homophony	.253	.06	.006
Induced GPC			
Baddeley Test	-.028	.00	ns
Meaning judgements	.243	.05	.016

involvement with rhyming suggests an involvement of the lexical-to-phonology route. To explain further, a high score on the rhyming task indicates a response interference effect. When a word pair (e.g., hive/give) has a similar string (*-ive*), there is a tendency to want to respond "same," but because they do not rhyme, the correct response is "different." Good readers can repress this interference effect better than poorer readers. Poorer readers are less able to reduce the salience of the printed form and concentrate on the phonological code that the string generates. Presumably good readers are more automatic readers and do not encounter this problem.

The second regression, with spelling as the dependent variable, produced a final regression with a multiple R of .51. This regression confirms the involvement of spelling ability with reading ability and effects of print exposure, as indicated in the first regression. The susceptibility to spelling regularity as dependent variable was examined in the third regression. The final regression in this case produced a multiple R of .41. The regularity variable tested the extent to which people are using the GPC route. It was found that this was associated with the alternating case variable and with homophony. This link showing that susceptibility to homophony is connected to being affected by regularity, is in agreement with Beech and Harris (1997) who found that spelling regularity and homophony were similar in effect when comparing hearing impaired young readers with their hearing counterparts, matched in reading level. The involvement of case alternation here suggests that those who have a more Gestalt-like representation of words, who are susceptible to interference by case alternation, are

TABLE 4
T-Test Comparisons on Key Variables for High Versus Low Scorers in Reading, Spelling, Regularity and Induced GPC.

	High scores			Low Scorers		
Variable	Mean	SD	N	Mean	SD	N
Good vs. Poor Readers						
Age	24.0*	8.6**	18	19.3	1.2	11
Author Recognition	30.6**	8.9	18	14.7	5.3	11
Semantic Link	62.2**	13.6	18	33.8	17.7	11
Spelling	65.3**	12.2	18	47.0	14.2	11
Good vs. Poor Spellers						
Author Recognition	28.9**	10.4*	22	16.8	6.2	21
Silent Adult Reading	78.0**	11.6	22	61.4	12.6	21
Meaning judgements	42.0*	9.1*	22	32.4	14.0	21
Most vs. Least Susceptible to Regularity						
Homophony	.12*	.17	8	-.11	.30	18
Alternating Case	-.01**	.11	8	-.28	.14	18
Good vs. Poor at Induced GPC						
Meaning judgements	45.8*	13.7	16	35.7	10.8	17

Note: Where SDs have an asterisk, this indicates a significant difference in variance and the appropriate t-test accounting for unequal variance was then computed. The level of significance is denoted by * for $p < .05$ and ** for $p < .01$.

also affected by homophony and spelling regularity. Finally, performance in the Induced GPC task was examined as the dependent variable. This was found to be connected only with the meaning judgement task. The final regression produced a multiple R of only .24 here. Thus those who were good at being able to read words by means of their GPC were also good at the meaning task. Despite having controlled for the effects of verbal reasoning by forcing in performance on the Baddeley task, one might argue that both these tasks require an element of problem solving, perhaps more than the other tasks.

A further set of analyses involved selecting extreme groups, by these same dependent variables, and examining them for significant differences. The purpose behind this was to remove the majority portion of participants and concentrate on those who were the most extreme on these dimensions. First, groups of good and poor readers, above and below one standard deviation from the mean, respectively, were selected and then compared on the rest of the key variables. The same exercise was then conducted on spelling, regularity and induced GPC and the results of all of these t test comparisons, where significant, are shown in Table 4.

The first analysis compared good readers (mean 87.3; SD 3.19) with poor readers (mean 47.4; SD 10.9) and confirmed the strong connections among ability in reading, knowledge of word meanings, spelling and exposure to print. It further showed that the good readers tended to be older than the poorer readers, showing the importance of frequency of exposure to print. The older one is, the more one will have developed the skill of reading. About 82 percent of the sample lay between 18 to 20 years of age, so there is a narrow range on this variable. If one views the learning curve as exponential and negatively accelerating, at this stage the increments will be relatively small. Nevertheless, this partly demonstrates that even during this three-year period there are noticeable improvements in reading skill. The mean age of 24 years for the good readers also suggests a strong influence of the 8 percent of older readers in this small sample. The first analysis also showed that good readers are better at working out problem-solving tasks that involved a reasonable knowledge of nuances of meaning. The second analysis compared good spellers (mean 73.4; SID 4.8) with poor spellers (mean 34.3; SD 3.8). It supported the importance of the reading-spelling-exposure triumvirate. In addition, it showed that good spellers were better at making meaning judgements. The meaning task has a strong element of requiring good spelling knowledge; so this result is unsurprising.

The last two analyses appear to be independent of considerations of reading or spelling ability, which also emerged from the multiple regression analyses. In the first, those most susceptible to regularity, in other words, those who are disrupted most by generating phonological codes at variance to a words actual code (as in *choir* vs. */kwIer/*) were compared to those least affected. As found in the multiple regression, they are affected by homophony and alternating case. They are also better at making meaning judgements. Finally, those who are good at doing the Induced GPC task (means 62.1 vs. 10.7) are also good at making meaning judgements, confirming the regression analysis.

The following main conclusions may be drawn on individual differences in mature readers in reading and spelling:

1. First, controlling for verbal reasoning, there was a not unexpected substantial "Matthew effect" (e.g., Stanovich, 1986), in that those who were good at reading were also good at spelling, good at working out the subtle manipulations of word meanings, and also good at recognizing authors. Similarly, there was a significant tendency for good readers to be older readers. The process of reading is a self-rewarding circle. Presumably good readers enjoy the process and in turn improve their reading skills, spelling skills, and are good at working out the subtle meanings of words. Awaida and Beech (1995) discuss a similar effect in a longitudinal study of young readers. Each year, the effects of being a good reader in the previous year had a significant impact on reading development. Explanations can range from motivational factors to the more simple relationship between practice and skill development. The Author Recognition Test supports this last explanation that suggests that good readers read more print and therefore become more practised or automatized.
2. Poor readers were more susceptible to response competition in a rhyming judgement task, according to the regression analysis. Thus, visually similar pairs (e.g., *fowl-bowl*) were more difficult to judge as not rhyming than

visually dissimilar pairs (e.g., *tight-slate*). Johnston and McDermott (1986) argued that in this task there is a need to recheck for phonological similarity for visually similar non-rhyming pairs, suggesting that poor readers have a greater propensity to do this in the current situation. Thus having generated the two different phonological codes for the two words, a checking operation on the letter strings interfered with the "different" response that they were going to make. A similar explanation is that poor readers were more likely to use their GPCs to identify the words as having the same phonological ending.

3. Independent of literacy skills, there were variations in the extent to which people used their GPCs, according to the regularity and homophony tasks. The conditions of the experiment were artificial in presenting sequences of unconnected single words. Under these circumstances the use of a GPC is more likely to be applied compared to reading simple text. The results showed that two different methods, regularity and homophony, produced similar effects, suggesting a common dimension. Case alternation was also implicated, more especially with regularity, indicating that those susceptible to using their GPC were more disrupted by case. We do not know at this point if this is due to disruption between or within the graphemic units.
4. These findings do not reveal a connection between poor GPC skills and poor lexical skills. This supports Olson et al.'s supposition of an independence in the aetiology of phonological versus lexical skills. Instead, the GPC route seems to be a construct that operates independently of level of literacy skills. Those weak in GPC effects did not appear to be poor readers (nor particularly good ones). However, the GPC does appear to have construct validity in terms of its mode of operation, in that homophony correlates with regularity. Interestingly, the Induced GPC task did not associate with these regularity and homophony tasks. This perhaps suggests a distinction between an automatically operating GPC route, as measured by regularity and homophony in mature readers, and a more conscious process that can be produced in the GPC Induction task.

NOTES

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Address correspondence to: John R. Beech, Psychology Department, University of Leicester, University Road, Leicester LE1 7RH, Leicestershire UK

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